

Data Visualization Dashboard Analysis

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Overview

This document explains the purpose and analysis of the data visualization dashboard. The dashboard presents a set of interactive visualizations to allow the user to explore different aspects of the data. It is designed to be intuitive, with a user-friendly interface that provides insights into crime incidents, trends, locations, and statistics.

The dashboard consists of several sections, each focusing on different aspects of the dataset. These include crime trends, district-level insights, crime types, and arrest rates. The goal is to provide actionable insights that can help in decision-making processes.

1 Dashboard Components and Features

1.1 1. Data Loading

Description:

- The function accepts an optional filename. If no filename is provided, it returns nothing.
- If the file is a valid CSV, it reads the content using pandas and returns the loaded data.

Requirements:

- The pandas library must be installed.
- The operating system library (os) is imported but not used in this function.

1.2 2. Data Processing Function

Description:

- The function removes duplicate rows and rows with missing values.
- Converts columns representing dates into datetime format.
- Ensures that certain columns expected to be integers are properly converted.
- Creates new time-related columns such as month name, year, month, day, hour, and weekday from the date.

- Generates a Unix timestamp from the date.
- Initializes a standard scaler and a label encoder, though only the label encoder is used.
- Encodes several categorical columns including primary type, description, location description, arrest status, and domestic cases.

Requirements:

- Pandas library
- Scikit-learn library (specifically StandardScaler and LabelEncoder)

1.3 3. Normalized Tables Creation Function

Purpose: This function organizes a DataFrame into several normalized tables, making the data more structured and relational for easier storage and analysis.

Description:

- A table for crime types is created by selecting primary type, description, and FBI code, removing duplicates, and assigning a unique crime type ID.
- A table for locations is created using columns like block, location description, coordinates, community area, ward, district, and beat, again assigning a unique location ID.
- A date table is created containing various time-related fields, with each unique combination assigned a unique date ID.
- The main DataFrame is updated by merging it with the crime types, locations, and date tables to include their respective IDs.
- Finally, a master incidents table is created, containing incident details along with the linked crime type ID, location ID, and date ID.

Outputs:

- A master incidents table.
- A crime types table.
- A locations table.
- A date table.

1.4 4. Read Tables from Database Function

Description:

- A connection to the specified database is created using SQLAlchemy.
- The connection is tested to ensure it is successfully established.

- Four tables are read from the database: MasterIncidents, Crime Types, Locations, and Dates.
- Each table is loaded into a separate pandas DataFrame, and a confirmation message is printed after each successful read.
- After reading the tables, the database connection is properly closed.

Outputs:

- Four pandas DataFrames representing the MasterIncidents, Crime Types, Locations, and Dates tables.

1.5 5. Streamlit App Setup and Usage Guide

5.1 Install Needed Libraries: The following Python libraries are required to run the app:

- Streamlit
- Pandas
- SQLAlchemy
- Scikit-learn
- Plotly

It is advisable to create a virtual environment first, but it is not necessary if you are just testing.

5.2 Saving App Code: Save your app's code in a Python file, such as `app.py`. This file will load the data, create graphs, and build the app layout.

2 Interactive Features of the Dashboard

The dashboard includes various interactive features for the users to explore the data. These features allow the user to filter, select, and visualize crime data from different perspectives:

2.1 1. Crime Type and Arrest Status

The dashboard includes a bar chart visualizing the distribution of crimes by type and arrest status. Users can interact with the chart by selecting different crime types and viewing how arrest rates vary across them.

2.2 2. Crime Trends over Time

A time series chart shows the trend of crime incidents over a period of time. The user can select the desired time range using a slider to explore crime data over different periods (7, 30, 90 days).

2.3 3. Crime Distribution by Location

A heatmap shows crime density by location, allowing the user to explore areas with high crime incidents. Users can zoom in and view more details based on specific regions and time periods.

2.4 4. Arrest and Domestic Crime Analysis

The dashboard displays visualizations comparing arrests vs. non-arrested incidents and domestic vs. non-domestic crimes. These visualizations are designed to help the user quickly understand patterns in arrest rates and domestic crime trends.

2.5 5. Filter Options

The sidebar includes several filtering options such as:

- Selecting datasets from different tables (e.g., Merged Dataset, Master Incidents, Crime Types).
- Filtering data by date range (using a start and end date).
- Selecting specific crime types.
- Filtering by crime count and arrest rate.

3 Conclusion

The dashboard offers a comprehensive and interactive way to analyze crime data, with a focus on incidents, arrests, and locations. It allows users to explore the data visually, apply filters, and derive meaningful insights from the information. The layout is designed to be user-friendly and intuitive, making it suitable for non-technical users as well.

This dashboard is a powerful tool for stakeholders looking to make data-driven decisions and gain insights into crime trends, locations, and arrest patterns.